# N A N O L I T E

Product Catalog - United States Vol 2.3

Nanolite LLC, USA · Nanolite LLC, UAE · Nanolite PTV LTD · Nanolite Enterprise · Nanolite Holdings LLC · Nanolite NGO

SINCE 2010

## PRODUCT CATALOG

WELCOME TO NANOLITE'S PRODUCT CATALOG, WHERE INNOVATION MEETS EXCELLENCE. EXPLORE OUR RANGE OF ADVANCED CONSTRUCTION SOLUTIONS DESIGNED TO ENHANCE EFFICIENCY, SUSTAINABILITY, AND PERFORMANCE. DISCOVER THE FUTURE OF BUILDING WITH NANOLITE.

## Introduction

At Nanolite, we are dedicated to revolutionizing the construction industry with our cutting-edge Cellular Lightweight Concrete (CLC) solutions. Our comprehensive range of products, from CLC blocks and panels to advanced machinery and production plants, is designed to meet the diverse needs of modern construction. With a focus on sustainability, efficiency, and superior performance, Nanolite products offer unparalleled advantages in building projects of all scales.

Explore our catalog to discover how our innovative materials and technologies can transform your construction process, reduce costs, and elevate the quality of your projects. From reducing dead loads to enhancing thermal and acoustic insulation, Nanolite products are engineered to deliver excellence and drive the future of sustainable construction. Join us on this journey towards a smarter, greener, and more efficient building industry.

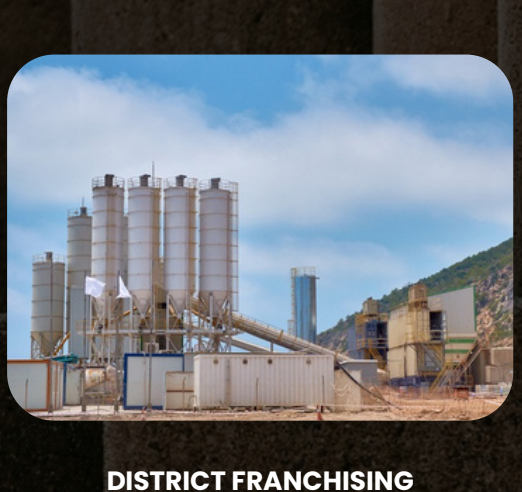
Discover the Future of Construction with Nanolite

- CLC Blocks & Panels: Lightweight, durable, and environmentally friendly solutions for building performance.
- Advanced Machinery: State-of-the-art equipment for efficient and reliable CLC production.
- Customized Solutions: Tailored products to meet specific project requirements, ensuring optimal results.

## OUR SERVICES

Welcome to the Nanolite Product Catalogue, your comprehensive guide to our innovative range of products designed to revolutionize the construction industry. At Nanolite, we are committed to delivering cutting-edge solutions that combine advanced technology with sustainable practices. Our product offerings are tailored to meet the diverse needs of modern construction, providing unparalleled performance, durability, and efficiency.

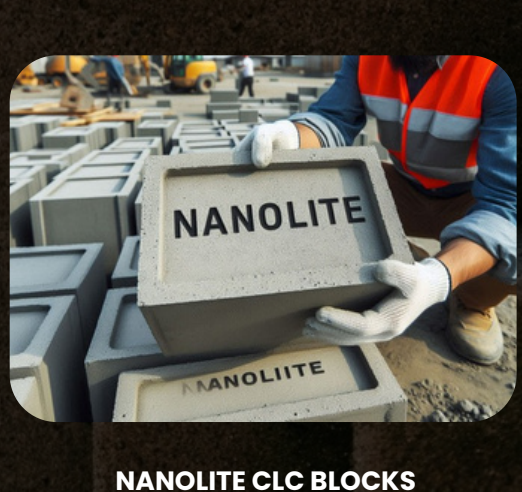
RESEARCH & DEVELOPMENT



DISTRICT FRANCHISING

NANOLITE EQUIPMENTS

BIM ENGINEERING



NANOLITE CLC BLOCKS

NANOLLITE READY MIX

NANOLITE PRECAST PANNELS

NANOLITE INSULATION POURING

Nanolite offers a diverse range of high-quality construction products designed to meet various needs:

- **Cellular Lightweight Concrete (CLC)** Blocks Plants: Advanced facilities for the production of lightweight, durable CLC blocks, which are ideal for a wide range of construction applications.
- **Light Weight Bricks Machines**: Equipment designed for the efficient production of lightweight bricks, offering superior insulation and structural benefits.
- **Fly Ash CLC Blocks Plants**: Specialized plants for producing CLC blocks using fly ash, contributing to sustainability and resource efficiency in construction.
- **CLC Blocks Making Equipment**: State-of-the-art machinery for the production of CLC blocks, ensuring precision, consistency, and high-quality output.
- **Foam Concrete Bricks Plants**: Facilities for manufacturing foam concrete bricks, known for their excellent thermal insulation and soundproofing properties.
- **CLC Light Weight Block Moulds**: High-quality moulds for creating CLC blocks with customizable sizes and specifications to meet specific project requirements.
- **Paver Block Plants and Machinery**: Equipment for the production of durable and aesthetically pleasing paver blocks for various landscaping and pavement applications.
- **Portable Foam Generators**: Compact and efficient generators for producing foam used in lightweight concrete applications.

At Nanolite, we blend innovative technology with a commitment to quality and customer satisfaction, delivering products and services that exceed expectations and drive success in the construction industry.

## NANOLITE CLC PRODUCTION BATCHING MACHINES

Nanolite offers a range of advanced machines designed to streamline the production of Cellular Lightweight Concrete (CLC) blocks and panels. Our machines are engineered for efficiency, precision, and durability, ensuring best & optimized results.

- Automation and Precision: Our machines feature automation systems that ensure precise mixing, batching, and cutting, enhancing consistency in every batch produced.
- Durability: Built with robust materials, Nanolite CLC machines are designed to withstand the rigors of high-volume production, offering long-term reliability.
- Ease of Operation: The user-friendly interface and centralized control panels make our machines easy to operate, even for those new to CLC production. With intuitive touch screen controls, remote access, managing operations is seamless.
- Customizable Solutions: From slurry mixers to foam mixers and cutting machines, our equipment is customizable to meet the specific needs of your production line.
- Efficiency: With options like the Automation Batching System and railway systems for block transport, our machines are designed to reduce labor costs, minimize waste, and speed up production cycles.

Nanolite CLC machines provide the foundation for high- quality, efficient, and scalable CLC production, making them an essential asset for any modern construction enterprise.

## Nanolite CLC Blocks

Nanolite CLC blocks are engineered to deliver exceptional performance for a range of construction applications. Lightweight and insulating, these blocks are designed for ease of handling and installation, making them an ideal choice for various building needs.

### Product Features:

- Customizable:Tailored to specific project requirements
- Exceptional Thermal Insulation: thermal conductivity of 0.09 W/mK
- Enhanced Acoustic Performance: Superior sound insulation
- Improved Fire Resistance: Withstands high temperatures Eco-
- Friendly: Manufactured with sustainable materials

### Technical Specifications:

Density Range: 800-2,000 kg/m³
Compressive Strength: 1-25 N/mm²
Water Absorption: 12-20%
Sound Insulation: 28-66 dB
Fire Resistance: Up to 4 hours

**Applications Usage :**

- High rise commercial buildings
- Lightweight retaining walls
- Insulating foundations
- Facade elements
- Residencial Villas
  Partition walls

## Nanolite Precast Pannels

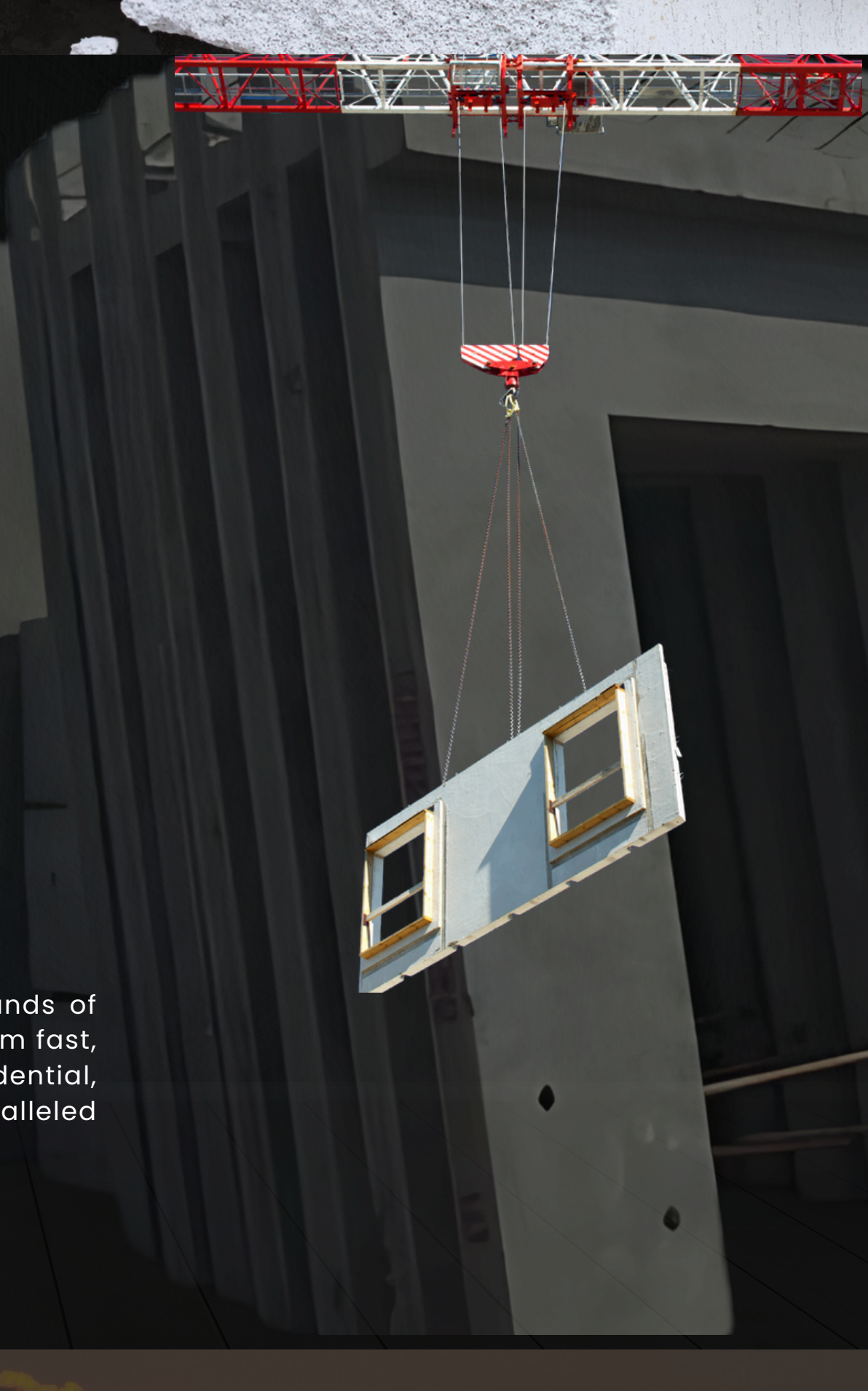
Nanolite’s precast panels offer a robust and efficient solution for large-scale construction projects, combining strength and durability with ease of installation. These panels are engineered to meet specific project requirements, making them ideal for a variety of applications.
Product Features:

- High Structural Integrity: Ensures safety and durability Fast
- Installation: Reduces construction time Customizable:
- Tailored to specific project requirements

Technical Specifications:

- Panel Thickness: Customizable from 50 mm to 200 mm
- Density: 1,000-2,000 kg/m³
- Compressive Strength: 5-25 N/mm²
- Thermal Conductivity: 0.09-0.59 W/mK
- Fire Resistance: Up to 4 hours

Nanolite Precast Panels are designed to meet the rigorous demands of contemporary construction, ensuring that each project benefits from fast, reliable, and high-quality building materials. Whether for residential, commercial, or industrial use, these panels deliver unparalleled performance and versatility.



## Comparison of Concrete Blocks and CLC Blocks

| Parameter | CLC Block | Red Brick | Hollow Concrete | AAC Blocks |
|---|---|---|---|---|
| Basic Raw Material and other inputs | Cement, Fly ash,silica fume, water, Foaming agent | Top soil of agricultural land and wood, coal, Bagasse and other fuel for firing Process in brick | Cement, sand, fly ash, lime and aggregate | Cement, Fly ash, Al Powder OH- Soloution |
| Production process | Plant / On-site Production 3-12/25-60/70- | kiln | Mobile or stationary plant | Plant / On-site Production |
| Compressive strength kg/cm2 | 400-2000 | 40-125 | 30-150 | 400-600 |
| Dry Density(kg/m³) | 200-2000 | 1800-2000 | 1900-2200 | 100-600 |
| Sound insulation | Superior | Poor | Better | Normal |
| Thermal Insulation | Superior | Poor | Good | Normal |
| Aging | Gains strength with age | No Gains strength with age | Yes | Slightly |
| Ease in working | Very easy, can be cut, nailed and drilled | Normal | Difficult | Very easy, can be cut, but not nailed |
| Eco Friendliness | Pollution free, Least energy requirement, consumes fly ash which is a waste from thermal power plant, Green building | creates smoke, Uses high energy for firing in kiln, Agricultural land becomes useless. | Low energy, No smoke | no smoke, low energy used Pollution free, |

### Comparison Analysis

When choosing between traditional concrete blocks and Cellular Lightweight Concrete (CLC) blocks, here are the key differences:

**Weight:** Concrete blocks are heavy, increasing structural load, while CLC blocks are lighter, reducing the need for reinforcement.
**Thermal Insulation:** Concrete blocks offer moderate insulation, whereas CLC blocks provide superior insulation, reducing energy costs.
**Acoustic Insulation:** CLC blocks excel in sound insulation, making them ideal for noise-sensitive areas.
**Fire Resistance:** CLC blocks offer enhanced fire resistance (up to 4 hours), outperforming concrete blocks.
**Eco-Friendliness:** CLC blocks are more sustainable, with a lower carbon footprint compared to concrete blocks.
**Handling and Installation**: CLC blocks are easier to handle and install, speeding up construction and lowering labor costs.
**Water Absorption:** CLC blocks absorb less water, reducing moisture-related issues.
**Cost Efficiency:** While initially more expensive, CLC blocks save costs in the long run due to reduced structural needs and faster installation.

## Nanolite CLC Ready Mix

Nanolite ReadyMixsolutions offer convenience and consistency for various construction needs. Designed for ease of use and high performance, Nanolite Ready Mix ensures quality and reliability in every batch.

### Product Features:

Consistent Quality: Ensured by advanced mixing processes
Versatile Applications: Suitable for various construction projects
High Strength and Durability: Meets industry standards
Technical Specifications:

- Density Range: 850-1,800 kg/m³
- Compressive Strength: 1-25 N/mm²
- Workability: High
- Consistency: Uniform across batches

Avalible in :

- 50 kg bags
- 1-ton bags

Lightweight · Water Resistant · Quick Construction · ECO Friendly · Sound Insulation

Fire Resistant · High Quality · Heat Insulation · High Tensile · Earthquake Resistant

**Nanolite Insulation Pouring**

Nanolite's insulationpouringsolutions are expertly designed to enhance the thermal and acoustic insulation of buildings. This method involves pouring a lightweight, insulating material into designated forms or cavities, offering flexibility and efficiency in application.
Product Features:

- **Excellent Insulation:** Provides superior thermal and acoustic insulation, improving indoor comfort and energy efficiency.
- **Flexible Application:** Easily adaptable to various forms and spaces, making it suitable for diverse construction needs.
- **Cost-Effective:** Reduces overall energy consumption and associated costs, offering long-term savings.

**Technical Specifications:**

- Density: 400-1,800 kg/m³
- Thermal Conductivity: 0.09-0.59 W/mK
- Water Absorption: 12-20%
- Sound Insulation: 28-66 dB

Fire Resistance: Up to 4 hours

**Applications:**

- Roof insulation
- Floor insulation
- Cavity filling
- Acoustic insulation

Nanolite Insulation Pouring solutions provide an effective and efficient way to enhance the energy performance and soundproofing of any building, ensuring long-lasting benefits and improved living conditions.

## MESSAGE FROM THE CHEIF EXECTIVE OFFICER - NANOLITE GROUP

At Nanolite Group of Companies, our journey began with a simple mission to revolutionize the way the world builds. What started with the innovation of high-performance Cellular Lightweight Concrete (CLC) blocks has now evolved into a multi-disciplinary group that spans construction, mining, aerospace, automation, and sustainable technologies. Every division of Nanolite carries the same DNA, innovation, impact, and integrity. Our flagship product, the Nanolite CLC block, was born out of years of R&D to create a lightweight, cost-effective, and eco-friendly building material. Today, it’s not just a product — it’s a movement toward smarter, faster, and greener construction.

To our partners, collaborators, investors, and clients — thank you for being part of this mission. As we scale globally from our roots in Pakistan to hubs in the United States and the UAE, our focus remains unshakable: to build a better world through purpose-driven innovation.

Warm regards,
Muhammad Momen
Cheif Exective Officer
Nanolite Group of Companies

FOUNDER & CEO – NANOLITE GROUP OF COMPANIES
Mr. Muhammad Momen

## MESSAGE FROM THE CHAIRMAN - NANOLITE GROUP

Over the years, Nanolite has grown into a trusted name in lightweight construction technology, pioneering innovations that bring together durability, affordability, and environmental responsibility. Our Lightweight Concrete Blocks and Ready Mix Materials stand as a testament to this commitment, already transforming projects across regions with unmatched quality and reliability.
But beyond products, Nanolite represents an idea the belief that construction can be lighter, smarter, and greener. We are proud to lead this movement, empowering partners, builders, and entrepreneurs through our franchise system, which opens doors for innovation and opportunity across the globe.

As we continue this journey, I extend my heartfelt gratitude to our dedicated team, valued partners, and all those who share our vision. Together, we are not just building structures. We are building the foundation for a better, more sustainable tomorrow.

With best regards,
Mr. Malik Mehmood
Chairman & Founder
Nanolite Group of Companies

CHAIRMEN – NANOLITE GROUP OF COMPANIES
Mr. Malik Mehmood

## CONTACT US

At Nanolite, we value your inquiries and are here to assist you with any questions or information you need about our CLC blocks and panels. Whether you’re interested in our products, need technical support, or want to explore franchise opportunities, our team is ready to help.

**CONTACT INFORMATION:**

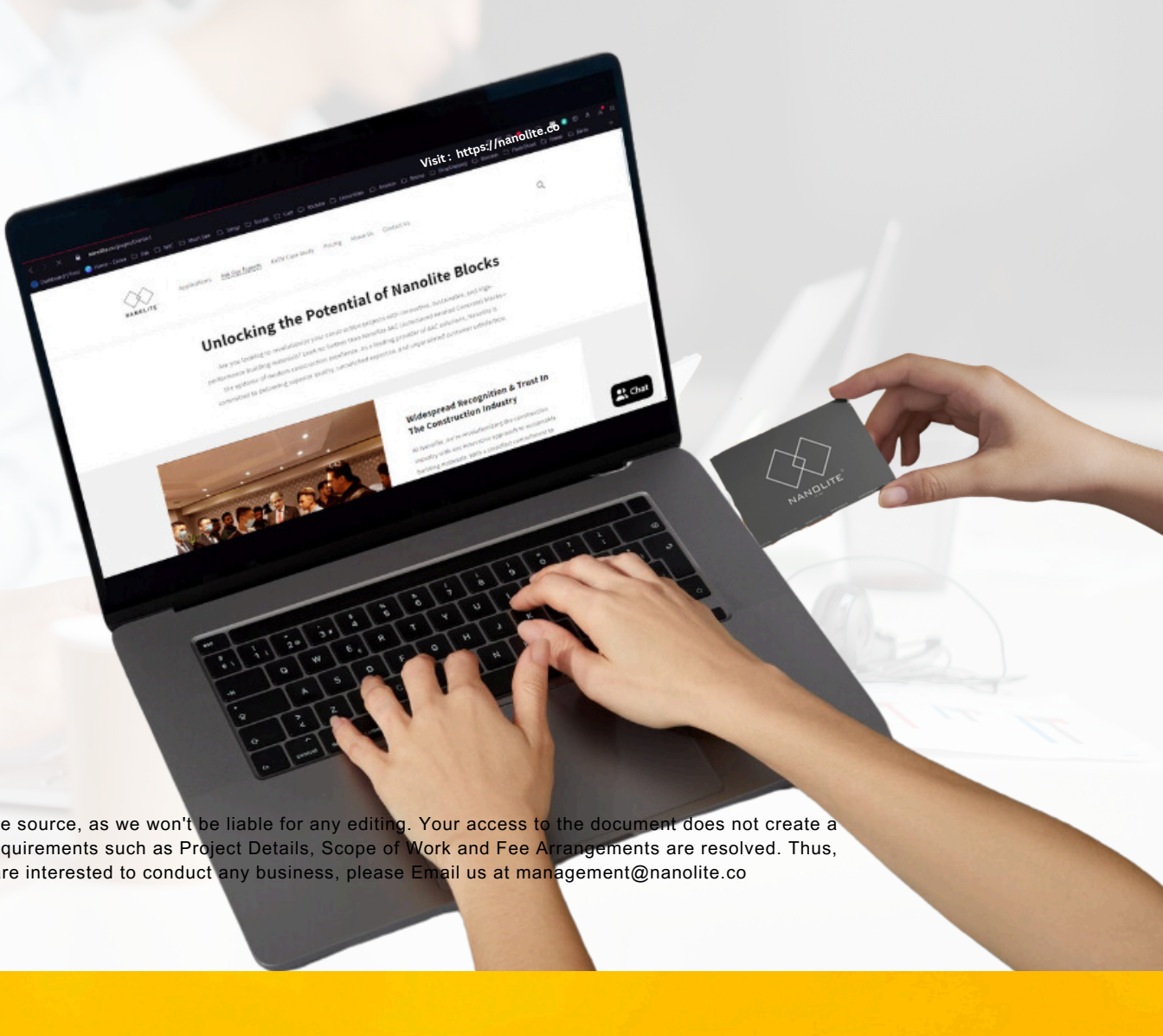
*For more information on franchise opportunities, please contact us:*
Address: 110-61 107 Ozone Park, Queens, Newyork. USA
Email: management@nanolite.co
Phone: +1 718 912 4171

**BUSINESS HOURS:**

Monday - Friday: 9:00 AM - 5:00 PM
Saturday / Sunday: Closed

**VISIT US:**

www.nanolitegroupofcompanies.com
www.nanolite.co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Widespread Recognition & Trust In The Construction Industry
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